

**Time:** 5 minutes**Outcome:** Determine equilibrium, shortages, surpluses, adjustment, and comparative statics.**Difficulty:** ●○○ Beginner

Simultaneous Shifts



THE CORE IDEA

When demand and supply shift at the same time, each shift contributes separately to the new equilibrium. Analyze how each change affects price and quantity, then combine the directional effects. If both shifts push one variable in the same direction, that result is determinate; if they push in opposite directions, the result depends on the relative size of the shifts.



HOW TO RECOGNIZE IT

- Both a demand determinant and a supply determinant change.
- A bicycle endorsement raises demand while higher input costs reduce supply.
- Analyze each shift separately before combining its price and quantity effects.
- Opposing effects on one variable leave its direction ambiguous unless relative shift sizes are known.

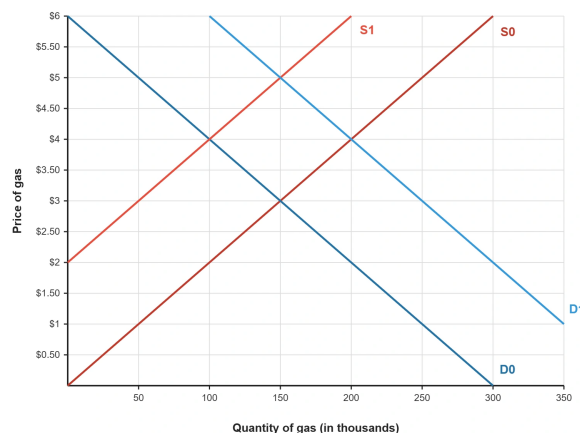


WATCH OUT

Do not assume an unchanged price means the market experienced no shifts. A demand increase raises price and quantity, while a supply increase lowers price and raises quantity. Equal price effects can cancel while quantity rises.



WORKED EXAMPLE: COMBINING DEMAND AND SUPPLY SHIFTS



The original curves D_0 and S_0 intersect at a price of \$3 and quantity of 150 thousand. After demand shifts right to D_1 and supply shifts left to S_1 , the new intersection is \$5 and the same quantity of 150 thousand. Both shifts raise price; their opposing quantity effects happen to offset in this graph, though quantity would otherwise be ambiguous.



CHECK YOURSELF

Which pair of market changes is shown when the graph moves from D_0/S_0 to D_1/S_1 ?

**READY?**

Return to the game and master this concept.